

HRMS Attendance Module

Comprehensive .NET Test Case Specification

Testing role: Senior .NET/API/SQL test specialist. This document covers positive, negative, boundary, validation, persistence, regression, integration, API and data-integrity tests for the current Attendance design.

Scope: Attendance processing, calendar classification, manual create/update, centralized attendance calculation, half-day leave integration, duplicate prevention, repositories, APIs, and required master/configuration data.

Expected test discipline: Execute positive and negative cases against a controlled test database. Verify both HTTP response and persisted database state where applicable.

1. Test Strategy

Use layered testing: unit tests for business rules/calculation, application-handler tests for orchestration, integration tests with SQL Server/EF Core repositories, API tests for controllers, and regression tests for existing Leave/Attendance behavior. For every calculated result, verify the derived columns—not only Status.

Key assertion fields: AttendanceDate, EmployeeId, AttendanceShiftId, AttendancePolicyId, CheckIn, CheckOut, WorkedMinutes, LateMinutes, EarlyLeaveMinutes, OvertimeMinutes, Status, Remarks, IsDeleted and audit fields where applicable.

2. Test Data / Preconditions

Required: valid EmployeeId; active EmployeeShiftAssignment effective on test date; active AttendanceShift; active AttendancePolicy; AttendanceDayStatuses WORKING_DAY/WEEKLY_OFF/HOLIDAY/LEAVE; raw attendance logs; CompanyHoliday data; LeaveRequestStatus APPROVED; approved LeaveRequest for leave tests.

Use at least two employees: one valid employee with assignment and one employee without assignment. Use normal, late, early-leave, overtime, overnight and missing-punch dates. Create separate test dates to avoid duplicate-record interference.

Important existing rule: Saturday/Sunday are weekly off. Half-day leave uses existing LeaveRequest.TotalDays = 0.50; full-day leave uses 1.00.

3. Test Case ID Convention

ATT-CAL = calculation; ATT-DAY = day-status/calendar; ATT-PROC = range processing; ATT-MAN = manual attendance; ATT-API = API; ATT-DATA = persistence/data integrity; ATT-NEG = negative/security/validation; ATT-REG = regression; ATT-PERF = performance.

4. Attendance Calculation — Positive & Boundary Cases

ID	Scenario / Input	Expected
ATT-CAL-001	CheckIn 09:00, CheckOut 18:00, Break 60	WorkedMinutes = 480; no late/early/overtime if policy matches.
ATT-CAL-002	CheckIn exactly at scheduled start	LateMinutes = 0; not Late.
ATT-CAL-003	CheckIn within grace period	LateMinutes = 0.
ATT-CAL-004	CheckIn exactly at grace boundary	LateMinutes = 0 because late condition is after threshold.
ATT-CAL-005	CheckIn one minute after grace threshold	LateMinutes > 0; Late status unless another higher-priority status applies.
ATT-CAL-006	CheckOut exactly at scheduled end	EarlyLeaveMinutes = 0.
ATT-CAL-007	CheckOut one minute before scheduled end	EarlyLeaveMinutes > 0; EarlyLeave unless another higher-priority status applies.
ATT-CAL-008	WorkedMinutes exactly FullDayMinutes	No overtime.
ATT-CAL-009	WorkedMinutes one minute above FullDayMinutes but below minimum overtime	OvertimeMinutes = 0.
ATT-CAL-010	WorkedMinutes exactly MinimumOvertimeMinutes above full day	Overtime awarded.
ATT-CAL-011	Overtime exceeds MaximumOvertimeMinutes	OvertimeMinutes capped at maximum.
ATT-CAL-012	Overtime disabled	OvertimeMinutes = 0 even when worked time is high.
ATT-CAL-013	CheckIn missing	Status MissingIn; derived minutes remain zero.
ATT-CAL-014	CheckOut missing	Status MissingOut; derived minutes remain zero.
ATT-CAL-015	CheckOut == CheckIn	Status MissingOut.
ATT-CAL-016	CheckOut before CheckIn	Status MissingOut.

ID	Scenario / Input	Expected
ATT-CAL-017	Elapsed time less than break	WorkedMinutes = 0; no negative minutes.
ATT-CAL-018	Normal attendance with no late/early/overtime	Status Present.
ATT-CAL-019	Worked minutes at half-day boundary	Verify exact configured half-day status boundary.
ATT-CAL-020	Worked minutes one minute below half-day boundary	Verify HalfDay.
ATT-CAL-021	Late + overtime	Status LateOvertime.
ATT-CAL-022	Late only	Status Late.
ATT-CAL-023	Early leave only	Status EarlyLeave.
ATT-CAL-024	Overtime only	Status Overtime.
ATT-CAL-025	Overnight shift 22:00–06:00	Scheduled end moves to next day; late calculation uses correct scheduled start.
ATT-CAL-026	Overnight shift with next-day checkout	No incorrect same-day scheduled-end calculation.
ATT-CAL-027	Zero worked minutes	Status Absent in centralized calculation.

5. Calendar / Day Status Tests

ID	Scenario	Expected
ATT-DAY-001	Saturday	WEEKLY_OFF.
ATT-DAY-002	Sunday	WEEKLY_OFF.
ATT-DAY-003	Monday with no holiday/leave	WORKING_DAY.
ATT-DAY-004	Active CompanyHoliday on date	HOLIDAY.
ATT-DAY-005	Deleted CompanyHoliday on date	Not treated as holiday.
ATT-DAY-006	Inactive CompanyHoliday on date	Not treated as holiday.
ATT-DAY-007	Approved full-day leave covering date, TotalDays 1.00	LEAVE; LeaveDays carried as 1.00.
ATT-DAY-008	Approved half-day leave covering date, TotalDays 0.50	LEAVE classification with LeaveDays 0.50; processor later writes HalfDay.
ATT-DAY-009	Pending leave covering date	Not treated as approved leave.
ATT-DAY-010	Rejected leave covering date	Not treated as approved leave.
ATT-DAY-011	Cancelled/deleted leave	Not treated as approved leave.
ATT-DAY-012	Approved leave starts on date	Leave detected.
ATT-DAY-013	Approved leave ends on date	Leave detected.
ATT-DAY-014	Date outside leave range	No leave.
ATT-DAY-015	APPROVED status inactive/deleted	No approved leave detected.
ATT-DAY-016	APPROVED status missing	No leave detected; verify expected configured-data behavior.
ATT-DAY-017	Date is both weekly off and holiday	Weekly off wins because it is checked first.
ATT-DAY-018	Date is holiday and approved leave	Holiday wins because holiday is checked before leave.
ATT-DAY-019	Date is weekly off and approved leave	Weekly off wins because it is checked first.

6. Process Attendance — Positive Cases

ID	Scenario	Expected
ATT-PROC-001	Process one normal working day with IN/OUT	One attendance record created with calculated values.
ATT-PROC-002	Process range with multiple working days	One record per eligible date.
ATT-PROC-003	Process range containing weekends	Weekly-off records created; no raw punch calculation required.
ATT-PROC-004	Process range containing holiday	Holiday calendar record created.
ATT-PROC-005	Process range containing full-day approved leave	LEAVE record created.
ATT-PROC-006	Process range containing approved half-day leave	HalfDay record created; no punches required.
ATT-PROC-007	Working day with no punches	Absent record created with no-punch remark.
ATT-PROC-008	Existing record for date	Date skipped; no duplicate.
ATT-PROC-009	Mixed month/range	Each date independently follows its calendar/punch rule.
ATT-PROC-010	Process after raw logs exist	First IN and last OUT are used.

ID	Scenario	Expected
ATT-PROC-011	Multiple IN/OUT punches	First IN and last OUT selected.
ATT-PROC-012	Created count	createdRecords equals actual records created, including applicable calendar records.

7. Process Attendance — Negative / Edge Cases

ID	Scenario	Expected
ATT-NEG-001	EmployeeId does not exist	No invalid record; controlled skip/error according to current handler contract.
ATT-NEG-002	Employee has no active assignment	Date skipped; no AttendanceRecord created.
ATT-NEG-003	Assignment effective after attendance date	No applicable assignment; date skipped.
ATT-NEG-004	Assignment expired before attendance date	No applicable assignment; date skipped.
ATT-NEG-005	Assignment deleted	Date skipped.
ATT-NEG-006	Assignment inactive	Date skipped.
ATT-NEG-007	Assigned shift missing/deleted/inactive	Date skipped.
ATT-NEG-008	Assigned policy missing/deleted/inactive	Date skipped.
ATT-NEG-009	AttendanceDayStatus master row missing	Controlled InvalidOperationException; verify API exception middleware response.
ATT-NEG-010	Duplicate existing record	Date skipped; database remains one non-deleted record.
ATT-NEG-011	Invalid FromDate > ToDate	Validator rejects request.
ATT-NEG-012	Range exceeds validator maximum	Validator rejects request.
ATT-NEG-013	Malformed GUID	API model validation returns 400.
ATT-NEG-014	Malformed DateOnly	API model validation returns 400.
ATT-NEG-015	Null required process body	API returns 400.

8. Manual Attendance Create Tests

ID	Scenario	Expected
ATT-MAN-001	Valid employee/date/check-in/check-out	201 Created and GUID returned.
ATT-MAN-002	Valid CheckIn only	Record created with MissingOut.
ATT-MAN-003	No CheckIn	Record created with MissingIn.
ATT-MAN-004	CheckOut <= CheckIn	Expected safe MissingOut behavior; verify no negative worked minutes.
ATT-MAN-005	Duplicate employee/date	ConflictException; API should return configured conflict response.
ATT-MAN-006	No assignment for date	NotFoundException; no record created.
ATT-MAN-007	Inactive/deleted shift	NotFoundException; no record created.
ATT-MAN-008	Inactive/deleted policy	NotFoundException; no record created.
ATT-MAN-009	Late check-in	LateMinutes calculated according to grace period.
ATT-MAN-010	Early check-out	EarlyLeaveMinutes calculated.
ATT-MAN-011	Overtime eligible	Overtime calculated according to policy.
ATT-MAN-012	Remarks supplied	Remarks persisted.
ATT-MAN-013	Null remarks	Record accepted if command validator permits null.

9. Update Attendance Tests

ID	Scenario	Expected
ATT-MAN-014	Update valid CheckIn/CheckOut	Derived fields recalculated.
ATT-MAN-015	Change Present to Late	LateMinutes/status recalculated.
ATT-MAN-016	Change CheckOut to early	EarlyLeave/status recalculated.
ATT-MAN-017	Change to overtime	Overtime/status recalculated.
ATT-MAN-018	Remove CheckIn	Status MissingIn.
ATT-MAN-019	Remove CheckOut	Status MissingOut.

ID	Scenario	Expected
ATT-MAN-020	Set CheckOut <= CheckIn	MissingOut; no negative minutes.
ATT-MAN-021	Non-existent record GUID	NotFoundException.
ATT-MAN-022	Deleted record GUID	NotFoundException.
ATT-MAN-023	Route id differs from command id	400 Bad Request; no update.
ATT-MAN-024	Change Remarks only	Remarks update; calculation remains consistent.
ATT-MAN-025	Persistence verification	Updated values are actually persisted after request completes.

10. API Contract Tests

ID	Endpoint / Input	Expected
ATT-API-001	POST /api/attendance-records/process valid body	200; message + createdRecords.
ATT-API-002	POST process invalid date range	400.
ATT-API-003	POST process missing body	400.
ATT-API-004	GET /api/attendance-records/{valid-guid}	200 AttendanceRecordDto.
ATT-API-005	GET with unknown GUID	404 according to exception middleware.
ATT-API-006	GET malformed GUID	400.
ATT-API-007	GET list default paging	200 PagedResult.
ATT-API-008	GET list employee filter	Only requested employee records.
ATT-API-009	GET list date filters	Only records inside requested range.
ATT-API-010	GET list status filter	Only requested status.
ATT-API-011	GET invalid page values	Validator/default behavior is respected; no server error.
ATT-API-012	POST /manual valid	201 + GUID.
ATT-API-013	POST /manual duplicate	Conflict response.
ATT-API-014	PUT valid id/body	204; persisted values updated.
ATT-API-015	PUT id mismatch	400.
ATT-API-016	PUT unknown GUID	404.

11. Database / Persistence Tests

ID	Check	Expected
ATT-DATA-001	Employee/date uniqueness behavior	No duplicate non-deleted attendance record for same employee/date.
ATT-DATA-002	Calendar record fields	Correct EmployeeId, ShiftId, PolicyId, AttendanceDate, Status and Remarks.
ATT-DATA-003	Half-day leave record	Status HalfDay; CheckIn/CheckOut null; minutes zero; approved half-day remark.
ATT-DATA-004	Full-day leave record	Status LEAVE; calendar remarks correct.
ATT-DATA-005	Absent record	Status Absent; CheckIn/CheckOut null; no-punch remark.
ATT-DATA-006	Working record	CheckIn/CheckOut and all calculated minute columns match calculation.
ATT-DATA-007	Soft-deleted record	Must not be treated as an active duplicate or approved leave.
ATT-DATA-008	Inactive master data	Not used for assignment/shift/policy/day status decisions.
ATT-DATA-009	Audit fields	Created/Updated audit behavior matches application conventions.
ATT-DATA-010	Transaction/error path	A failed date should not leave a partially malformed AttendanceRecord.

12. Regression Tests

ID	Regression	Expected
ATT-REG-001	Existing Present attendance	Still Present.

ID	Regression	Expected
ATT-REG-002	Existing EarlyLeave	Still EarlyLeave and minutes correct.
ATT-REG-003	Existing Overtime	Still Overtime and minutes correct.
ATT-REG-004	Existing WeeklyOff records	Still WEEKLY_OFF.
ATT-REG-005	Existing Holiday records	Still HOLIDAY.
ATT-REG-006	Existing Absent behavior	Still Absent when no punches.
ATT-REG-007	Existing full-day leave	Still LEAVE after half-day enhancement.
ATT-REG-008	Manual create	Still calculates correctly.
ATT-REG-009	Manual update	Still recalculates correctly.
ATT-REG-010	Existing GET APIs	Response schemas unchanged.
ATT-REG-011	Existing paging/filtering	Results remain correct.
ATT-REG-012	Build	dotnet build succeeds with no errors/warnings that indicate a regression.

13. Security / Robustness Tests

ID	Scenario	Expected
ATT-NEG-016	Unauthorized request when authentication is required by deployment	401.
ATT-NEG-017	Authenticated user without required authorization, if policies are later applied	403.
ATT-NEG-018	SQL injection strings in status/filter parameters	No SQL injection; parameterized query behavior.
ATT-NEG-019	Extremely long Remarks	Validation/database limits enforced; no 500.
ATT-NEG-020	Concurrent processing same employee/date	No duplicate active records; uniqueness/idempotency behavior verified.
ATT-NEG-021	CancellationToken cancellation	Operation stops cleanly without corrupting data.
ATT-NEG-022	Database unavailable	Controlled error response; no partial corruption.
ATT-NEG-023	Missing master configuration	Controlled error/skip as defined; no malformed record.

14. Unit Test Recommendations for .NET

Use xUnit/NUnit plus Moq (or the project's existing mocking framework). Unit-test AttendanceCalculationService without SQL Server. Mock repositories for AttendanceCalendarBusinessRules. Handler tests should verify repository calls, skip behavior, created count and exception paths. API integration tests should use the project's real DI and persistence configuration where possible.

High-value unit tests: IsWeeklyOff for Saturday/Sunday/weekday; holiday active/inactive/deleted; approved leave status/date boundaries; worked-minute formula; grace boundary; overtime minimum/maximum; missing punches; overnight schedule; status precedence; half-day LeaveRequest.TotalDays mapping.

Avoid testing only implementation details. Assert business outcomes and persisted values.

15. Acceptance Criteria

- ✓ All positive calculation cases pass.
- ✓ All negative/validation cases return controlled outcomes.
- ✓ Weekly off, holiday and leave precedence are correct.
- ✓ Full-day approved leave remains LEAVE.
- ✓ Half-day approved leave TotalDays 0.50 becomes HalfDay.
- ✓ No duplicate active attendance record is created for employee/date.
- ✓ Manual Create and Update recalculate derived fields consistently.
- ✓ Process range returns correct createdRecords.
- ✓ API contracts remain backward compatible.
- ✓ SQL persistence matches calculated values.
- ✓ Regression suite passes for previously verified attendance scenarios.
- ✓ dotnet build succeeds.

16. Final Execution Order

1) Run build. 2) Run unit calculation tests. 3) Run calendar/Leave integration tests. 4) Run process-range tests. 5) Run manual Create/Update tests. 6) Run API contract tests. 7) Verify SQL rows. 8) Run full regression. 9) Run concurrency/robustness tests. 10) Record evidence: request, response, database row, logs and test result.

Important: Some negative-case expected HTTP status codes depend on the project's `ExceptionHandlerMiddleware` mapping. Where the current source does not explicitly define the mapping, the test should verify the configured middleware response rather than assuming a particular status code.